

Conditional units and force paths

Exact algebra checked. Physical specimen calibration remains missing.

1 Supply physical normalizers

$$C = l_{ref} \times F_{scale} \quad [N]$$

$$\ell = L_{ref} / \Gamma \quad [m]$$

Γ is dimensionless; $\gamma = 2 \text{ Ns dps}$ is a length.

Missing paired force / area / tare
and reference length ledger

2 Convert one compatible circuit

$$\text{Smooth PE prefactor: } C \text{ kpe} / K \quad [N]$$

$$\text{PE width: } \ell / K \quad [m]; \text{ slope: } C \text{ kpe} / \ell \quad [N/m]$$

$$\text{SE prefactor: } C \text{ kse0} \quad [N]; \text{ exponent: } \text{kse} / \ell \quad [1/m]$$

Smooth PE branch only
Published parity unresolved

3 Declare area and force topology

$$\text{Parallel capacity: } T = H \text{ kCB dps } Q$$

Serial count: adds length / energy; same force

$$\text{Cauchy} = (\lambda / J) \times \text{nominal stress}$$

PCSA vs projected FCSA:
cosine applied once

What the audit establishes

12 exact identities • 4 intended dimensional rejections

2 unit-consistent mechanical errors exposed: serial-force multiplication and double pennation projection

No empirical calibration, kinetic ODE, tissue solve or convergence comparison.